Assignment 5 Report

Financial Engineering — Structured Products

Group 5
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Abstract

This report presents the results for Assignment 5 of the Financial Engineering course. We cover the calibration of the LMM caplet spot-volatility surface and the pricing of a 10y structured bond with embedded caplet and digital optionality (Exercise 1, points a–g), and the pricing of an exotic cap on Libor differences under the Bond Market Model via Monte Carlo (Exercise 2). The structured-bond exercise also addresses fine and coarse-grained Delta-bucket sensitivities, total Vega, Delta hedging with par swaps, Vega hedging with ATM caps and digital risk correction via call-spread replication. The exotic cap is calibrated caplet-by-caplet from the LMM stripping into BMM bond vols and priced under the spot measure with a full-dimensional Cholesky-correlated Monte Carlo: all 16 forward bonds are jointly evolved on the reset grid, exploiting the Markov-chain-on-reset-dates property of BMM, and the spot-measure stochastic discount is built path-by-path as the running product of the bonds at fixing.

Exercise 1: Structured bond pricing and risk

On 15 February 2008, Bank XX issues a 10-year structured bond with notional $N = €50\,\text{Mln}$, Start Date 19 February 2008. The two legs of the swap hedging structure are:

- **Party A (Bank XX)** pays quarterly Euribor 3m + 2% (Act/360, Modified Following);
- **Party B (I.B.)** pays $X\%\cdot N$ at the Start Date and structured coupons thereafter: a fixed 4% first quarter, then a capped Libor-plus-spread payoff with digital jumps, sub-divided in three blocks (up to 3y, 3y–6y, after 6y).

The fair upfront $X\%$ closes the gap between the two legs at the Start Date.

Point a) LMM Caplet Spot Volatility Stripping

The market quotes flat cap volatilities on a grid of maturities $T_y \in \{1, 2, \ldots, 12, 15, 20\}$y and 13 strikes from 1.50% to 10.00%. From these quotes we extract the underlying *caplet spot volatilities* using a recursive bootstrapping procedure.

The resulting surface is shown in Figure 1 and sampled on the cap-maturity grid in Table 1.

Table 1 reports the calibrated LMM caplet spot volatilities sampled on the cap-maturity grid. For each maturity Y, we report the spot volatility of the caplet that *fixes* at year Y and pays at $Y + 0.25$y (i.e. row index $4Y$ of the full 79×13 surface).

Mat	1.50	1.75	2.00	2.25	2.50	3.00	3.50	4.00	5.00	6.00	7.00	8.00	10.00
1Y	16.12	14.69	14.06	13.63	15.19	17.20	19.05	19.99	20.03	21.18	22.18	23.40	25.49
2Y	23.97	21.34	18.94	18.92	20.68	24.62	28.06	29.55	26.45	26.32	27.23	28.06	29.51
3Y	27.11	25.04	22.62	21.06	20.42	17.97	16.03	16.04	20.41	22.59	24.00	25.19	26.81
4Y	23.44	22.16	21.24	20.83	20.50	20.49	20.51	20.07	19.25	20.15	21.41	22.52	24.61
5Y	24.89	23.29	21.92	20.95	20.07	17.77	15.89	15.42	16.92	18.37	19.88	20.85	22.49
6Y	23.68	22.46	21.58	20.81	19.53	18.39	17.31	16.30	15.67	16.72	17.76	18.75	20.69
7Y	23.41	22.21	21.30	20.45	19.86	17.49	15.58	14.55	15.06	15.73	16.80	17.96	19.31
8Y	23.11	22.11	21.26	20.05	19.16	17.13	15.88	15.03	14.02	15.16	15.91	16.78	18.37
9Y	22.82	21.70	20.63	20.02	18.53	16.76	14.78	13.82	14.09	14.76	15.58	16.21	17.73
10Y	22.42	20.84	20.60	19.58	18.44	17.11	15.54	15.18	13.34	12.83	14.00	15.17	16.09
12Y	21.33	20.15	18.97	18.18	18.15	15.78	14.99	12.51	13.51	14.10	13.78	13.96	14.96
15Y	20.79	20.23	19.40	18.42	16.24	15.25	13.64	13.64	11.83	11.69	12.87	13.69	14.77
20Y†	20.52	18.17	16.95	16.64	16.96	14.68	13.11	11.27	11.79	12.04	12.10	12.65	13.92

Table 1: LMM caplet spot volatilities (%) sampled on the cap-maturity grid. Each entry is the spot volatility of the caplet *fixing* at year Y (paying at $Y+0.25$y), extracted from the full 79×13 bootstrapped surface. Strikes in %. † The stripping grid has 79 caplets whose last reset falls at 19.75y; for the nominal 20y row we therefore report the spot vol of the caplet fixing at 19.75y (paying at 20y), the closest available point on the grid.

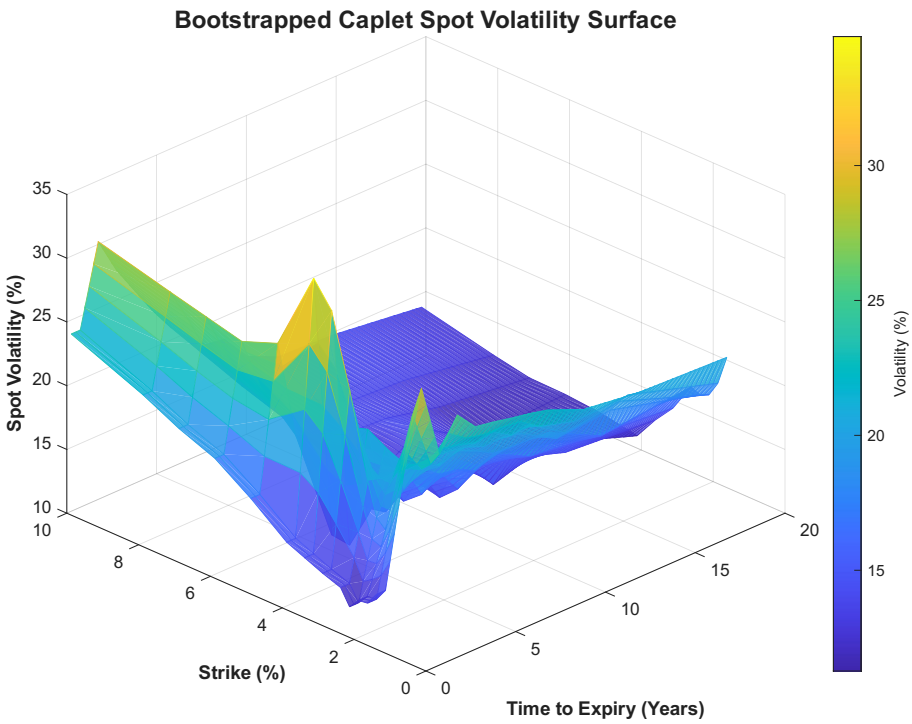

Figure 1: Bootstrapped Caplet Spot Volatility Surface (79×13). Time-to-expiry on x-axis, strike on y-axis, spot vol on z-axis.

b) Upfront X^* — Standard Pricing

The fair upfront payment X^* is determined by imposing $\mathrm{NPV}_A = \mathrm{NPV}_B$. Using the single-curve framework, we decompose the two legs:

Party A Leg (Floating) The Net Present Value (NPV) is calculated as:

$$\mathrm{NPV}_A = N\left(1 - B(0,T)\right) + N \cdot s \cdot \mathrm{BPV}_{\text{float}}$$

where $s = 2.00\%$ is the contractual spread and $\mathrm{BPV}_{\text{float}}$ is the Basis Point Value of the floating leg.

Party B Leg (Structured Coupons) The structured coupon stream is decomposed into vanilla components to allow for independent pricing. The first quarter is a deterministic 4.00% coupon. For quarters $k = 2, \ldots, 40$, the payoffs are split into three sub-periods, each represented as a combination of a floating rate, a short Caplet, and a short Digital option:

$$\begin{aligned}
\text{[Up to 3y]} \quad & \text{Cpn}_k = L + 1.00\% - (L - 4.20\%)^+ - 0.70\% \, \mathbf{1}_{L>4.20\%} \\
\text{[3y – 6y]} \quad & \text{Cpn}_k = L + 1.20\% - (L - 4.70\%)^+ - 1.00\% \, \mathbf{1}_{L>4.70\%} \\
\text{[After 6y]} \quad & \text{Cpn}_k = L + 1.30\% - (L - 5.40\%)^+ - 1.10\% \, \mathbf{1}_{L>5.40\%}
\end{aligned}$$

Pricing Methodology:

- **Caplets**: Priced using the Black-76 formula.
- **Digital Options**: Priced via the closed-form Black digital formula.
- **Volatility Calibration**: For each maturity T_k, the volatility σ is obtained by cubic spline interpolation on the strikes of the calibrated LMM spot-volatility matrix.

Results and Upfront Calculation The fair upfront X^* is the value that equalizes the two legs, expressed as a percentage of the notional N:

Quantity	**Value**
NPV_A (Party A Floating Leg)	$26,064,888.09$ EUR
NPV_B (Party B Coupons Leg)	$19,475,961.77$ EUR
Fair Upfront X^*	**13.1779%**
Upfront Amount ($X^* \cdot N$)	$6,588,926.32$ EUR

Table 2: Fair upfront calculation under standard Black digital pricing.

The decomposition reveals that Party B is effectively paying Euribor plus a spread while structurally selling caps and digital options to Party A. The "digital jump" represents the gap between the floating rate and the fixed trigger level, which significantly influences the total upfront value.

Point c) Delta-bucket sensitivities

We compute the Delta-bucket DV01 of the structured bond by bumping each quoted bootstrap instrument by +1 bp and re-pricing the bond after recalibrating the spot vols. For each instrument we re-bootstrap the curve via the `shift`-struct interface of `bootstrap.m`, recompute the caplet maturities with the new discount factors, re-price the bond with updated spot vols, and compute the Bank XX MTM change as

$$\text{DV01}_k^{XX} = -(X_k^* - X^*) \cdot N$$

since for Bank XX the locked $X^* \cdot N$ remains fixed while the fair X^* moves under the bump.

A selection of the most sensitive instruments is shown in Table 3; the dominant contribution is the 10y swap, which fixes the longest discount factor $B(0, 10\text{y})$ entering both legs.

Instrument	**Mat (y)**	ΔX^* **(% of** N**)**	**DV01**^{XX} **(EUR)**
Depo_3	0.079	+0.010630	−5,315.15
Fut_1	0.332	+0.000480	−218.98
Fut_7	1.825	+0.001019	−509.32
Swap_2y	2.003	−0.010296	5,148.01
Swap_4y	4.005	−0.002973	1,486.72
Swap_5y	5.005	−0.003002	1,501.21
Swap_6y	6.005	+0.004039	−2,019.57
Swap_7y	7.005	−0.002516	1,257.75
Swap_10y	10.008	+0.058361	−29,180.44
TOTAL (sum of fine-grained DV01)			−26,735.33
Parallel +1 bp shift (sanity check)			−23,994.11
Relative gap			11.42%

Table 3: Point (c) – Delta-bucket sensitivities (selected rows). Total over all 60 quoted instruments vs. parallel-shift sanity check.

The $\sim 11\%$ gap between the sum of fine-grained DV01 and the parallel-shift DV01 is genuine second-order non-linearity. The bond contains many caplet and digital options whose payoff is non-linear in the curve, so a sequence of small bumps does not yield exactly the same MTM change as a single parallel bump.

d) Total Vega

To quantify the sensitivity of the structured bond to changes in market volatility, we perform a total Vega calculation. Every cell of the flat-volatility surface is bumped by +1% in absolute terms. Following the shift, the LMM spot-volatility matrix is re-calibrated, and the bond is re-priced while keeping the discount curve constant. The resulting change in Mark-to-Market (MtM) is:

$$\boxed{\text{Vega}^{\text{XX}} = -142500.2847 \text{ EUR per } +1\% \text{ shift in the flat-vol surface}}$$

The negative sign indicates that Bank XX is **net short volatility**. Through the structured coupon features—specifically the caps on Libor and the digital jumps—Bank XX has implicitly sold optionality to the counterparty (I.B.). Consequently, a parallel increase in volatility increases the market value of the options held by I.B., leading to a corresponding loss for Bank XX.

e) Delta Hedging with 2y, 6y, and 10y Par Swaps

To manage the portfolio's interest rate exposure, we aggregate the fine-grained DV01 sensitivities into three coarse-grained buckets (0–2y, 2–6y, 6–10y) using *triangular weights* across the year fractions. The three weight vectors are defined as:

$$\begin{aligned}
w_{0\text{-}2}(\tau) &= \mathbf{1}_{\tau\le 2} + \frac{6-\tau}{4}\mathbf{1}_{2<\tau\le 6},\\
w_{2\text{-}6}(\tau) &= \frac{\tau-2}{4}\mathbf{1}_{2<\tau\le 6} + \frac{10-\tau}{4}\mathbf{1}_{6<\tau\le 10},\\
w_{6\text{-}10}(\tau) &= \frac{\tau-6}{4}\mathbf{1}_{6<\tau\le 10} + \mathbf{1}_{\tau\ge 10}.
\end{aligned}$$

These weights sum to 1 everywhere. Operationally, each bucket DV01 is computed by re-bootstrapping the curve with a shift vector equal to w_b times +1 bp, and re-pricing the bond, after having re-calibrated the spot volatilities. We find the following bucketed DV01s:

Coarse Bucket	**DV01^{bond} (EUR)**
0–2y	90.29
2–6y	2 643.98
6–10y	−26 896.53
Sum	−24 162.26
Parallel +1 bp (sanity check)	−23 994.11

Table 4: Coarse-grained DV01 of the structured bond using triangular weights.

The 6–10y bucket dominates the risk profile by an order of magnitude, which is consistent with the 10-year term structure of the bond.

To completely hedge the Delta risk, we select three par swap notionals $(N_{2y}, N_{6y}, N_{10y})$ such that the net DV01 in each coarse bucket is zero. Let M be the 3×3 sensitivity matrix where:

$$M_{ij} = \text{DV01 of swap } j \text{ on coarse bucket } i.$$

We solve the linear system $M\,\mathbf{N} = -\mathbf{DV01}^{\text{bond}}$.

Why start with the longest swap? A 2y swap has no cash flows beyond year 2, meaning its DV01 in the 6–10y bucket is exactly zero. Similarly, a 6y swap has virtually no sensitivity in the 6–10y bucket. Consequently, the matrix M is upper-triangular.

To solve the system, we *must* proceed backwards:

1. The **10y swap** is the *only* instrument capable of neutralizing the massive exposure in the 6–10y bucket. Its notional is determined first.
2. The **6y swap** is then sized to neutralize the 2–6y bucket, accounting for the "spillover" DV01 generated by the 10y swap in this intermediate sector.
3. Finally, the **2y swap** absorbs the residual risk in the 0–2y bucket.

Solving this upper-triangular system yields the following hedging notionals:

Hedging Swap	**Notional (EUR)**
2y par swap	−699 192.85
6y par swap	−5 274 562.01
10y par swap	+34 022 173.36

Table 5: Notional amounts of the par swaps required for full Delta hedging.

f) Vega Hedging with 6y and 10y ATM Caps

To manage volatility risk, we aggregate the bond's Vega into two coarse buckets (0-6y and 6-10y) using triangular weights on the flat-vol maturities and the same bump-and-revalue scheme of point (d) restricted to each weight profile. The resulting bucketed sensitivities are:

$$\text{Vega}^{\text{bond}}_{\text{0-6}y} = -15,953\,\text{EUR}, \qquad \text{Vega}^{\text{bond}}_{\text{6-10}y} = -125,190\,\text{EUR}.$$

We apply the same shocks to two ATM caps (6y and 10y) with strikes equal to their respective ATM swap rates to construct the following Vega sensitivity matrix:

$$\Sigma_{\text{caps}} = \begin{pmatrix} \text{Vega}^{\text{cap6y}}_{\text{0-6}y} & \text{Vega}^{\text{cap10y}}_{\text{0-6}y} \\ \text{Vega}^{\text{cap6y}}_{\text{6-10}y} & \text{Vega}^{\text{cap10y}}_{\text{6-10}y} \end{pmatrix} = \begin{pmatrix} 1.30 \cdot 10^{-3} & -1.89 \cdot 10^{-5} \\ 0 & 2.69 \cdot 10^{-3} \end{pmatrix}.$$

The matrix Σ_{caps} is upper-triangular by construction. Specifically, $\Sigma_{\text{caps}}(2,1) = 0$ because a 6y cap does not contain any caplets resetting beyond the 6-year mark, making it unable to influence the long-end bucket.

This structure indicates that the hedging process must begin with the 10y cap, as it is the only available instrument capable of neutralizing the risk in the 6-10y bucket. Once the 10y cap notional is fixed, the 6y cap is sized to absorb the remaining exposure in the 0-6y bucket, including any secondary effects caused by the 10y instrument. Solving the linear system $\Sigma_{\text{caps}}\mathbf{N} = -\mathbf{Vega}^{\text{bond}}$ (via MATLAB's backslash) yields the following notionals:

Hedging Cap	**Notional (EUR)**
6y ATM Cap	$+13,216,295$
10y ATM Cap	$+46,517,255$

Table 6: Notional amounts for the ATM caps required for full Vega neutralization.

The positive notionals indicate that Bank XX is short volatility on the structured bond and must purchase caps to neutralize the position.

g) Digital Risk and Call-Spread Replication

The discontinuous payoff of the structured bond introduces significant digital risk. To account for the volatility smile, we replace the standard Black-76 digital pricer with a **call-spread** replication using a width of $\varepsilon = 1$ bp:

$$\widehat{\text{Digital}}(K) \approx \frac{C(K, \sigma(K)) - C(K+\varepsilon, \sigma(K+\varepsilon))}{\varepsilon} \cdot \text{Gap}$$

- **Caplet Pricing**: $C(K, \sigma(K))$ is the Black-76 caplet price at strike K.
- **Volatility**: Both $\sigma(K)$ and $\sigma(K+\varepsilon)$ are interpolated from the calibrated spot-vol surface to capture the local smile slope.
- **Replication**: The formula converges to the smile-adjusted price $(-\partial C/\partial K)$ as $\varepsilon \to 0$.

By applying this correction to the digital components identified in point (b), we obtain the following results:

Pricing Scheme	X^*	$X^* \cdot N$ **(EUR)**
Black Digital (point b)	13.1779%	$6,588,926.32$
Call-Spread Replication (point g)	12.9446%	$6,472,281.17$
Difference	-23.3 bp	$-116,645.15$

Table 7: Digital risk correction via call-spread replication.

The corrected upfront X^* is lower than the Black-76 estimate. This occurs because the negative slope of the volatility surface increases the market cost of the digital coupons paid by Party B. As the value of the coupon leg increases, the required balancing upfront payment decreases. The magnitude of the correction (23 bps in X^*, ∼117,000 EUR on the notional) is non-trivial, on the order of the bid-ask of the structure itself, and ignoring it leads to mispricing.

Exercise 2: Exotic cap under BMM

Payoff and pricing measure

The exotic cap pays quarterly at reset T_{i+1} the amount

$$\delta(T_i, T_{i+1}) \left[L(T_i, T_{i+1}) - L(T_{i-1}, T_i) - 5\,\text{bp} \right]^+$$

for $i = 1, \ldots, 15$, with maturity 4y and first payment 6m after the Start Date. The first "previous" Libor $L(T_0, T_1)$ is the spot Eur 3m at t_0, deterministic.

We work under the *Bond Market Model* (BMM) with the spot-measure dynamics:

$$dB_i(t) = B_i(t)\, v_i(t) \cdot \Big[-\rho \sum_{j=k+1}^{i-1} v_j(t)\, dt \; - \; dW(t) \Big], \qquad T_k < t \le T_{k+1},$$

together with the fixing assumption $v_i(t) = 0$ for $t \ge T_i$ and the deterministic Markov-chain-on-resets structure for the discount factor:

$$D(t_0, T_{i+1}) \;=\; \prod_{j=0}^{i} B_j(T_j),$$

where $B_j(t) := B(t; T_{j-1}, T_j)$ is the forward ZCB for the j-th period.

Calibration of $v_i^{\mathbf{BMM}}$

For each caplet i we calibrate v_i^{BMM} at the assignment-required strike $K_i = L(t_0; T_i, T_{i+1}) = F_i(0)$ (ATM-forward) by matching the two pricing formulas:

1. **Black-Libor (LMM)** – the price of the same caplet with the LMM stripped vol σ_i^{LMM} obtained by spline interpolation on strike at K_i of `spot_vols_matrix`$(i, :)$, the already bootstrapped spot volatilities:

$$C_i^{LMM} = \delta_i\, B(0, T_{i+1}) \left[F_i(0)\, \Phi(d_1^L) - K_i\, \Phi(d_2^L) \right].$$

2. **Black-Bond (BMM)** – the equivalent representation of the caplet as a put on the forward ZCB $B_{i+1}(t)$ with strike $B^* = 1/(1 + K_i \delta_i)$:

$$C_i^{BMM} = (1 + K_i \delta_i)\, B(0, T_i) \left[B^*\, \Phi(-d_2^B) - B_{i+1}(0)\, \Phi(-d_1^B) \right],$$

   where $d_{1,2}^B$ use $v_{i+1}^{\text{BMM}} \sqrt{T_i}$.

We solve $C_i^{LMM} = C_i^{BMM}$ for v_{i+1}^{BMM} via `fzero`, using the low-rates approximation $v \approx \sigma^L \delta F/(1+\delta F)$ as initial guess. The calibrated values are reported in Table 8; they are very small (ten to thirty bp), as expected from the relation $v \approx \sigma^L (1 - B_{\text{fwd}})$ that maps Libor vols to bond-price vols.

i	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
v_i ($\times 10^{-3}$)	1.5	1.4	1.5	2.0	2.6	3.1	0.9	2.7	2.3	1.8	1.4	1.5	1.8	2.1	2.2

Table 8: Exercise 2 – Calibrated BMM bond vols at strike $K_i = F_i(0)$.

Spot-measure simulation

We work under the spot measure. For the exotic option, the price of each coupon is

$$\mathrm{Cpn}_i(0) \;=\; \mathbb{E}^{\mathrm{spot}}\Big[D(t_0,T_{i+1})\cdot\delta_i\big(L_{\mathrm{new}}-L_{\mathrm{prev}}-K\big)^+\Big],$$

with $L_{\mathrm{new}} = L(T_i,T_{i+1})$, $L_{\mathrm{prev}} = L(T_{i-1},T_i)$, $K = 5$ bp, and the path-wise stochastic discount $D(t_0,T_{i+1}) = \prod_{j=0}^{i} B_j(T_j)$ built from the bond values frozen at their own resets.

The simulation jointly evolves all $n+1 = 16$ forward bonds $B_0, B_1, \ldots, B_{15}$ in a single Monte Carlo pass:

1. **Correlation set-up.** We build the full 16×16 correlation matrix $\rho_{ij} = \exp(-\lambda\,|T_i - T_j|)$ on the Act/365 reset times and compute its Cholesky factor L, $\rho = LL^\top$. The factor is rebuilt on each step on the sub-block of alive bonds to keep the dimensions tight.
2. **Time stepping.** The simulation walks the reset grid $T_0 = 0 \to T_1 \to \cdots \to T_{16}$, with one bond freezing at each step. At step k, every bond $i \geq k$ (still alive) is evolved over the interval $(T_{k-1}, T_k]$ with the BMM step-vol exact lognormal update:

$$\log B_i(T_k) = \log B_i(T_{k-1}) - \tfrac{1}{2}v_i^2\Delta_k - \underbrace{v_i \sum_{\substack{j \text{ alive}\\ j\neq i}} \rho_{ij}\, v_j\, \Delta_k}_{\text{Girsanov drift}} - v_i\sqrt{\Delta_k}\,\widetilde{Z}_i,$$

   where $\widetilde{Z}_i = (L\mathbf{Z})_i$ are correlated normals built from i.i.d. $\mathbf{Z} \sim \mathcal{N}(0,I)$ via the Cholesky factor L of the alive-block correlation matrix.
3. **Freezing and discounting.** At each reset T_k, $B_k(T_k)$ becomes deterministic and is multiplied into the running cumulative discount $D(t_0,T_k) = \prod_{j=0}^{k-1} B_j(T_j)$ (BMM Markov-chain identity).
4. **Coupons.** Once the simulation reaches T_{16}, the values $B_j(T_j)$ for $j = 0,1,\ldots,15$ are available on every path. The Libor at fixing is recovered as $L_j(T_j) = (1 - B_j(T_j))/(\delta_j\, B_j(T_j))$, and the i-th coupon PV is the path-wise average of $D(t_0,T_{i+1})\cdot\delta_i\,(L_i(T_i) - L_{i-1}(T_{i-1}) - 5\,\mathrm{bp})^+$, with the discount built incrementally as $D(t_0,T_{i+1}) = \prod_{j=0}^{i} B_j(T_j)$. All 15 coupons are priced from the same set of paths: this captures both the correlation between the two Libors entering the same coupon and the cross-coupon dependence through the shared discount factors.

The seed is fixed (`rng(42)`) for reproducibility.

Result

Quantity	**Value**
Price (% of notional)	1.1292%
Monte Carlo std error	3.68×10^{-6} (0.033% of price)
Monte Carlo paths	10^6
Random seed	42
Price on $N = 50$ Mln	$\approx$ 564,600 EUR

Table 9: Exercise 2 – Exotic cap price under BMM, spot-measure Monte Carlo with full Cholesky correlation across all 16 forward bonds.

The price is small relative to the notional ($\sim$ 113 bp) for two reasons: the difference $L_{\mathrm{new}} - L_{\mathrm{prev}}$ between adjacent Libors has very small effective volatility because the two underlyings are highly correlated ($\rho_{i,i+1} = e^{-0.1\cdot 0.25} \approx 0.975$); and the threshold of 5 bp on top of the difference makes the coupon active only when the curve forwards re-price meaningfully upward between two consecutive quarters, an event of moderate frequency.

Problems and Criticalities

Choice of measure for Exercise 2. A first attempt at the simulation worked under the T_{i+1}-forward measure with a Girsanov drift correction on B_i, treating each coupon independently with a bivariate (B_i, B_{i+1}) Monte Carlo. We dropped this approach for two reasons. First, under the T_{i+1}-forward measure neither $B_i(t)$ nor $B_{i+1}(t)$ is a martingale (the natural martingale measure for $B_j(t) = B(t, T_j)/B(t, T_{j-1})$ is the T_{j-1}-forward, not T_j-forward), so both bonds carry a Girsanov drift. Second, this scheme neglects the correlation between non-adjacent bonds and fails to consistently reuse the same path of frozen $B_j(T_j)$ across coupons, underpricing the structure. The spot-measure full multidimensional approach used here captures all ρ_{ij} correlations, and reuses one set of paths for all coupons.

Additional Functions Implemented

- `[dates, rates] = readExcelData(filename, formatData)`
- `[dates, rates] = readExcelData_MAC(filename, formatData)`
- `capVols = readCapVols(filename, sheetName)`
- `[dates, discounts, zeroRates] = bootstrap(datesSet, ratesSet, shift)`
- `zero_rates = from_discount_factors_to_zero_rates(reference_date, dates, discounts)`
- `disc_int = get_discount_factor_by_zero_rates_linear_interp(reference_date, interp_date, dates, discounts)`
- `adjusted_date = business_date_offset(base_date, varargin)`
- `spot_vols = bootstrap_vol(flat_vols_at_strike, strike, spot_vol_parameters, cap_maturities)`
- `spot_vols_matrix = compute_spot_vols_Eur_3m(flat_vols, spot_vol_parameters)`
- `output = compute_caplets_maturities(flat_vols, dates_bootstrap, discounts)`
- `[spot_vols_cap_grid, sampled_T_expiry, is_fallback] = spot_vols_on_cap_grid(spot_vols_matrix, flat_vols, spot_vol_parameters)`
- `plot_spot_vol_surface(spot_vols_matrix, flat_vols, spot_vol_parameters, filename)`
- `print_spot_vols_table(vols_matrix, flat_vols, label, is_fallback)`
- `upfront = compute_upfront(notional, spot_vols_matrix, strike_grid, spot_vol_parameters, dates_bootstrap, discounts, start_date, maturity_date_unadj, first_coupon_rate, mode_after_6y)`
- `BPV = BasisPointValueFloating(ref_date, maturity_date, dates, discounts)`
- `price_coupon = compute_price_coupon_up_to_3years(fwd_libor, spot_vol_caplet, spot_vol_digital, df_payment, T_expiry, yf_caplet)`
- `price_coupon = compute_price_coupon_3y_to_6y(fwd_libor, spot_vol_caplet, spot_vol_digital, df_payment, T_expiry, yf_caplet)`

- price_coupon = compute_price_coupon_after_6y(fwd_libor, spot_vol_caplet, spot_vol_digital, df_payment, T_expiry, yf_caplet, mode)
- digital_cs_value = compute_digital_call_spread(fwd_libor, T_expiry, df_payment, yf_caplet, K, epsilon, gap, spot_vols_matrix, strike_grid, caplet_idx)
- upfront = compute_upfront_dig_risk(notional, spot_vols_matrix, strike_grid, spot_vol_param, dates, discs, start_date, mat_date_unadj, first_coup_rate, mode_after_6y)
- price_coupon = compute_price_coupon_up_to_3years_dig_risk(fwd_libor, spot_vol_caplet, df_payment, T_expiry, yf_caplet, spot_vols_matrix, strike_grid, caplet_idx)
- price_coupon = compute_price_coupon_3y_to_6y_dig_risk(fwd_libor, spot_vol_caplet, df_payment, T_expiry, yf_caplet, spot_vols_matrix, strike_grid, caplet_idx)
- price_coupon = compute_price_coupon_after_6y_dig_risk(fwd_libor, spot_vol_caplet, df_payment, T_expiry, yf_caplet, mode, spot_vols_matrix, strike_grid, caplet_idx)
- bucket = compute_DV01_buckets(datesSet, ratesSet, strike_grid, flat_vols, notional, start_date, mat_date_unadj, first_coupon_rate, mode_after_6y, X)
- coarse = coarse_DV01_triangular(datesSet, rSet, flat_vols, spot_vols_mat, K_grid, notional, st_date, mat_date_unadj, first_coup_r, mode_after_6y, X)
- W = triangular_weights_swap(yf)
- delta_NPV = compute_delta_NPV_swap(rSet, dates, discs, st_date, coarse)
- notional_swaps = compute_portfolio_hedged_with_swap(coarse, delta_NPV)
- vega = compute_total_vega(dates, discounts, flat_vols, notional, st_date, mat_date_unadj, first_coupon_rate, mode_after_6y, spot_vol_params, X)
- output = coarse_vega(dates, discounts, flat_vols, notional,st_date, mat_date_unadj, first_coupon_rate, mode_after_6y, spot_vol_parameters, X)
- output_vega = compute_st_bond_coarse_vega(market_dates, discount_factors, flat_vol_data, bond_notional, st_date, mat_date, initial_fixed_rate, payoff_mode, calibration_params, X)
- bucket_weights = triangular_weights_vega(years_to_maturity)
- vega_matrix = compute_vega_sensitivity_matrix(flat_vols, ratesSet, spot_vols_base, output_vega, dates, discounts)
- DV01_cap = compute_delta_NPV_cap(flat_vols, ratesSet, spot_vols_matrix, output, dates, discounts)
- hedging_notionals = compute_vega_hedge_notionals(st_bond_vega_results, vega_sensitivity_matrix)
- notional_caps = compute_portfolio_hedged_with_cap(output, DV01_cap)

- `v_bmm = calibrate_bmm_vols(spot_vols_matrix, strike_grid, spot_vol_parameters, n_caplets)`
- `price = caplet_price_bmm(K, B_pay, B_reset, delta, T, v)`
- `[price, std_err] = price_exotic_cap_mc(v_bmm, spot_vol_parameters, lambda, n_caplets, spread, n_paths)`
- `cap_price = calculate_cap_price(spot_vol_mat, target_K, params, mat_lim, strike_grid)`
- `p = cap_increment_price(sigma_right, sigma_left, T_knots, T_to_interp, fwd, K, r, T, yf)`